

2.1.18 Heads

Read / write heads refer to the main reading and writing apparatus in a hard disk. The Read head contains thin films that display the GMR effect. The Write head is usually made of an electromagnetic substance that gets magnetised when electricity is passed to it, thus magnetising the spot below it.

2.1.19 Hybrid Drives

These are hard drives that include both rotating disks as well as Flash memory based storage. This is an attempt to combine the benefits of the two types of storage media: the inexpensiveness of rotating platters and the non-volatility of Flash memory. The Flash memory component in a hybrid drive is merely a temporary storage area; the data is finally written onto the disk. Like the cache in regular hard drives, the Flash memory improves performance, and since unlike cache memory the data is not lost when powered down, startup time is reduced since there is no longer the need to wait for the disks to rev up.

2.1.20 Landing Zone

During normal operation, the read/write heads levitate over the platter surface thanks to the air cushion created by the rotation. When powered down, the heads are brought to rest over an area on the platter where data is not stored. This is the landing zone.

2.1.21 LBA

Logical Block Addressing refers to the scheme used by modern hard disks to overcome the bottlenecks presented by older BIOS limitations that do not detect large-capacity hard disks. LBA allocates a unique number to each sector. The hard disk circuitry translates this into the respective CHS values, which is comprehended by the BIOS.

2.1.22 Logical Drives

This refers to the partitions created in an Extended Partition. A maximum of 24 logical drives can be created, since each drive needs to be lettered, and letters A and B are reserved for floppy disks.